Guangdong Technion
Israel Institute of Technology
广东以色列理工学院

Calculus 1 - 104003

Exam A
January 18, 2021

Your ID Number:

First name and Surname:

Guidelines

1. **Duration: 3 hours**. Use of calculators, personal dictionaries, electronic devices, reference materials, personal notes or any other extra material is not allowed.
2. Explain your solutions, quote theorems you are using.
   No credit will be given for non-justified answers!.
   Write clear and complete answers for each problem in the same page.

1. (a) Find all the pairs of real numbers a and b that satisfy simultaneously (8 p)

$$\lim_{x\to 0}\frac{\sqrt{ax^2+bx+1}-1}{x}=3 \qquad \text{and} \qquad \lim_{x\to\infty}\frac{\sqrt{ax^2+bx+1}-1}{x}=2.$$

(b) Consider

$$\lim_{x\to 0}\frac{x^2\sin\frac{1}{x}}{\sin x}$$

i. Is it possible to apply L'Hospital rule in order to compute this limit? (4p)

ii. Compute this limit. (8 p)

Answer

2. Consider the functions $f(x) = x^3 + 2$ and $g(x) = 2x^2 + 2$.

   (a) Prove that the graphs of these functions intersect in two points. (5 p)

   (b) Describe the tangent lines for both curves at these points. Are they equal? (10 p)

   Answer

3. (a) Let $f(x) = x^3 + x$. Compute $(f^{-1})'(0)$ and $(f^{-1})'(2)$. (7p)

   (b) Let $f, g : \mathbb{R} \to \mathbb{R}$ such that $f(g(x^2 + x)) + 3g(x) = 3\tan(x) + 1$, $g(0) = 0$ and $g'(0) = 2$. Compute $f'(0)$. (8 p)

Answer

4. (a) Find the extremes (local and global) of the following function

$$f(x) = \begin{cases} 2x^2 - 2x - 2 & \text{if } x \leq 2 \\ x & \text{if } x > 2 \end{cases}$$

and make the graph. (8p)

(b) Let $f : \mathbb{R} \to \mathbb{R}$ be a function that is differentiable in its domain and satisfies the following conditions:

i. $f'(-1) = f'(-\frac{1}{2}) = f'(0) = f'(\frac{3}{2}) = 0$;

ii. $\{x \in \mathbb{R} : f'(x) > 0\} = (-\infty, -1) \cup (0, \frac{3}{2})$;

iii. $\{x \in \mathbb{R} : f'(x) < 0\} = (-1, -\frac{1}{2}) \cup (-\frac{1}{2}, 0) \cup (\frac{3}{2}, +\infty)$.

Determine the local extremes for this function. Justify. Make a graph of a function satisfying these conditions. (7p)

<u>**Answer**</u>

5. (a) Find a differentiable function $f(x)$ such that f has a critical point at $x = 2$, the point $(2, 3)$ belongs to the graph of f and $f'(x) = -x + a$ for some real number a. (7p)

   (b) Compute $\int_3^9 x\sqrt{x+1}dx$. (10p)

Answer

6. Prove that

(a) $\sum_{n=0}^{\infty} \frac{2^n+3^n}{6^n} = \frac{7}{2}$ and $\sum_{n=1}^{\infty} \frac{2^n+3^n}{6^n} = \frac{3}{2}$. (6p)

(b) The real number $0.444\ldots$ is a rational number. (6p)

(c) $\sum_{n=1}^{\infty} \frac{1}{n2^n}$ converges. (6p)